

Question number	Answer	Additional guidance	Mark
2(a)(i)	An explanation that includes the following points: - because we do not want {other bacteria in the culture / contamination of culture} (1) - as they could {give an over-estimate of the number of bacteria / compete for nutrients / produce toxins (to the bacteria being cultured) / as they could be pathogens} (1) OR - because we do not want to transfer the bacteria onto {ourselves / surroundings} (1) - as they may be {pathogenic / harmful} (1)	ACCEPT microorganisms / pathogens ACCEPT to ensure only the test bacteria are used competition for space ACCEPT prevent infection / contamination of person NB Prevent infection with pathogenic bacteria = 2 marks	(2)

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2(a)(ii)	An explanation that includes the following points: • technique described (1) • technique explained (1)	ACCEPT pathogens / microorganisms / spores e.g. working by a bunsen burner / safety hood + to create an updraft / to move the microorganisms in the air away / to prevent bacteria falling into culture IGNORE kill bacteria in the air autoclaving {agar / culture fluid / equipment} using sterile equipment disinfecting work area washing hands + to kill any bacteria present reducing exposure of cultures to air / keep lid on plates / close {doors / windows} + to reduce bacteria falling into cultures	(2)

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2(b)	• 3.2 (1) • 3 / 2.7 / 2.66 / 2.658 (1)	ACCEPT $\log 5.2 - \log 2 = 3.2$ ECF 0.3 / 0.34 / 0.345 = 1 mark Bald answer of {3 / 2.7 / 2.66 / 2.658} = 2 marks Bald answer of 2.6578073089700996677 {incorrectly rounded up to three decimal places max / correctly rounded to more than three decimal places} = 1 mark Bald answer of 3.2 = 1 mark Bald answer of {0.3 / 0.34 / 0.345} = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
2(c)(i)	• 63 / 63.1 (: 1) OR 0.02 / 0.016 (:1)		(1)

Question number	Answer	Additional guidance	Mark
2(c)(ii)	An answer that includes the following points: • (student P) {cell counts (not using an exclusion dye) / optical methods / turbidity / absorption of light / colorimeter / counting chamber} as total cell counts (1) • (student Q) {dilution plating / colony counting / counting chamber using an exclusion dye} as only living cells counted (1)	NB both mark points but the wrong way round for 1 mark ACCEPT haemocytometer / Neubauer chamber living and dead IGNORE calorimeter ACCEPT {haemocytometer / Neubauer chamber} using an exclusion dye named exclusion dye e.g. eosin / trypan blue	(2)